

# Bootstrap tests for mean value differences over fuzzy samples

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**Abstract:** This paper aims to statistically test the null hypothesis  $H_0$  for equality of the mean values of 1D continuous parameter in two different populations, each represented by a fuzzy sample of i.i.d. observations. A degree of membership to the corresponding population is assigned to any of the observations in the fuzzy sample. A Bootstrap algorithm is developed for simulation-based approximation for the CDF of the difference between the sample means, provided that  $H_0$  is true. The needed  $p_{value}$  of the statistical test is derived using the constructed conditional distribution of the test statistic. The main idea of the proposed Bootstrap test is that, if  $H_0$  is true, then each sample can be shifted so that to have mean value equal to the fuzzy mean value of the united fuzzy sample. The two shifted fuzzy samples are summarized into conditional sample distributions of the 1D continuous parameter used for generation of synthetic couples of fuzzy samples in different pseudo-realities. The proposed algorithm has eight modifications, which differ by the method to generate the synthetic fuzzy sample, by the type of the conditional sample distributions derived from the shifted fuzzy samples, and by the type of the statistical test performed (one-tail or two-tail test). Numerical example are included to demonstrate the proposed algorithm.

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## 1. INTRODUCTION

Assume that there are two 1D (one-dimensional) samples of a continuous parameter that contains respectively  $n_1$  and  $n_2$  observations. Assume also that the observations  $z_k^1$  and  $z_k^2$  respectively of the first and of the second sample belong to two populations, called Population 1 and Population 2, with degrees of membership respectively  $\mu_k^1$  and  $\mu_k^2$ . In that sense, the data has fuzzy interpretation, hence we name the samples as Fuzzy Sample 1 and Fuzzy Sample 2:

$$Z^j = \left\{ \left( z_1^j - \mu_1^j \right), \left( z_2^j - \mu_2^j \right), \dots, \left( z_{n_j}^j - \mu_{n_j}^j \right) \right\}, \text{ for } j=1,2 \quad (1)$$

The objective is to define how the different condition in which the fuzzy samples were measured influenced the mean values of the distributions of the continuous 1D parameter in Population 1 and in Population 2. This may also be interpreted as the case where the two fuzzy samples were formed by two different populations. Then it is necessary to test the possibility that Population 1 has the same mean value as Population 2.

Most statistical tests that deal with the equality of the numerical characteristic  $C$  of the distribution laws of the random variables (r.v.) in Populations 1 and 2, calculate the value  $s$  of a given estimator  $S$  of the difference between the numerical characteristic estimates  $\hat{c}_1$  and  $\hat{c}_2$ , derived from the samples (1). Then they assess how typical  $s$  is under the

assumption that Populations 1 and 2 have equal values of the numerical characteristic  $C$ , i.e.  $c_1=c_2$ . The estimator  $S$  is called test statistic, and is a random variable whose values are such that the higher the difference between the true values  $c_1$  and  $c_2$  the higher the generated value. It is obvious that the formula to calculate  $s$  for the specific samples implicitly depends on the procedure to calculate the numerical characteristics  $\hat{c}_1$  and  $\hat{c}_2$ .

Let  $c_1$  and  $c_2$  be the true values of the mean value  $C$  respectively for Population 1 and 2.

This paper discusses a Bootstrap statistical test with eight modifications, which analyze differences in the mean values of two populations. Each modification has a null hypothesis that the two populations have equal mean values. The objective is to calculate  $p_{value}$ , which shows the probability to observe a difference between the fuzzy sample mean values not less than the one that was measured, provided that  $H_0$  is true. Algorithms on how to calculate  $p_{value}$  of those Bootstrap modifications shall be presented.

## 2. FUZZY SAMPLE MEAN VALUES AND STATISTICAL HYPOTHESES

Let's calculate the mean values of the observations in the two fuzzy samples as follows:

$$\hat{c}_j = \sum_{k=1}^{n_j} \mu_k^j z_k^j \Big/ \sum_{k=1}^{n_j} \mu_k^j, \text{ for } j=1,2 \quad (2)$$

Let's call  $\hat{c}_1$  and  $\hat{c}_2$  *fuzzy sample means*. It is only natural to calculate the fuzzy sample mean of the unified fuzzy sample, which unites the observations in (1) for  $j=1$  and for  $j=2$  as:

$$\hat{c} = \left( \hat{c}_1 \sum_{k=1}^{n_1} \mu_k^1 + \hat{c}_2 \sum_{k=1}^{n_2} \mu_k^2 \right) \Big/ \left( \sum_{k=1}^{n_1} \mu_k^1 + \sum_{k=1}^{n_2} \mu_k^2 \right) \quad (3)$$

As mentioned earlier, the null hypothesis  $H_0$  for all tests is “the mean values of the two populations are equal:  $c_1=c_2$ ”. Then the quantity

$$\Delta_{1-2,r}^{mean} = \hat{c}_1 - \hat{c}_2 \quad (4)$$

shall be almost equal to zero. Formula (4) takes into account the fact that the observations in the samples  $Z^1$  and  $Z^2$  are random, hence  $\Delta_{1-2}^{mean}$  is a random variable, and  $\Delta_{1-2,r}^{mean}$  is a possible realization. Each “substantial” deviation of  $\Delta_{1-2,r}^{mean}$  from zero casts doubt over the validity of  $H_0$ . Therefore  $\Delta_{1-2}^{mean}$  can be used as a test statistic for equality of the mean values.

The alternative hypotheses  $H_1$  vary across the different tests. In all two-sample Bootstrap tests for mean values, the  $H_1$  is that “the mean value of the first population is different from that of the second population:  $c_1 \neq c_2$ ”. In all one-tail tests,  $H_1$  depends on the calculated mean values in the two samples. If Fuzzy Sample 1 has a higher fuzzy sample mean than Fuzzy Sample 2, then the test is one-tailed of first type and  $H_1$  is “the mean value of Population 1 is higher than that of Population 2:  $c_1 > c_2$ ”. If Fuzzy Sample 1 has a smaller fuzzy sample mean than Fuzzy Sample 2, then the test is one-tailed of second type and  $H_1$  is “the mean value of Population 1 is smaller than that of Population 2:  $c_1 < c_2$ ”. In the rare case, where Fuzzy Sample 1 has the same sample mean value as Fuzzy Sample 2, then the one-tail tests are not performed.

### 3. BOOTSTRAP APPROACH TO ESTIMATE THE CONDITIONAL DISTRIBUTION OF THE TEST STATISTIC

Bootstrap simulation is one of the most reliable procedures to calculate the conditional distribution of the statistic  $S$  if  $H_0$  is true [Efron, Tibshirani, 1993]. The only assumption here is that the observations in the empirical samples are i.i.d. The Bootstrap simulation is a computer intensive method since it needs a large number of calculations [Press et al., 2007; Politis, 1998].

The work [Nikolova, Mihaylova, Tenekedjiev, 2013] presents a Bootstrap procedure to construct the conditional probability distribution of the test statistic  $\Delta_{1-2}^{mean}$  if  $H_0$  is true for the case when each observation in the samples (1) belong with complete certainty to the respective Population. In this paper, the ideas from the aforementioned work are generalized into a numerical simulation algorithm to calculate  $p_{value}$  of the

statistical tests for equality of the mean values of the two populations for the case of fuzzy samples as in (1).

Assume that the continuous cumulative distribution functions (CDFs) of the two populations  $CDF_{tr,1}(\cdot)$  and  $CDF_{tr,2}(\cdot)$  have the same mean values. Then the true value of  $\Delta_{1-2}^{mean}$  shall be

$\Delta_{1-2}^{mean}=0$ . These two distributions are not available to the analyst, but are realized statistically by the pair of samples  $(Z^1 - Z^2)$ , given in (1). Using (4) over these samples, it is possible to calculate the real estimate  $\Delta_{1-2,r}^{mean}$  of the test statistic.  $CDF_{tr,1}(\cdot)$  and  $CDF_{tr,2}(\cdot)$  would generate countless number of *hypothetical* sample pairs *similar* to  $(Z^1 - Z^2)$ :  $(Z_1^1 - Z_1^2)$ ,  $(Z_2^1 - Z_2^2)$ ,  $(Z_3^1 - Z_3^2)$ , ... . The analyst could

have calculated the hypothetical realizations  $\Delta_{1-2,1}^{mean}$ ,  $\Delta_{1-2,2}^{mean}$ ,  $\Delta_{1-2,3}^{mean}$ , ... of the test statistic based on each of the paired hypothetical samples. The hypothetical realizations  $\Delta_{1-2,1}^{mean}$ ,  $\Delta_{1-2,2}^{mean}$ ,  $\Delta_{1-2,3}^{mean}$ , ... would have revealed the probability distribution of the sample numerical characteristic  $\Delta_{1-2}^{mean}$ , and  $\Delta_{1-2,r}^{mean}$  would be one possible realization. The difference  $\Delta_{1-2}^{mean} - \Delta_{1-2,r}^{mean} = \Delta_{1-2}^{mean} - 0 = \Delta_{1-2}^{mean}$  is a random variable, called *root* and its distribution entirely describes whether the real test statistic  $\Delta_{1-2,r}^{mean}$  is different from zero by chance. The main task of the simulation techniques is to describe the distribution of *root* if  $CDF_{tr,1}(\cdot)$ ,  $CDF_{tr,2}(\cdot)$ , and the hypothetical sample pairs  $(Z_1^1 - Z_1^2)$ ,  $(Z_2^1 - Z_2^2)$ ,  $(Z_3^1 - Z_3^2)$ , ... are not known. Since the true  $CDF_{tr,1}(\cdot)$  and  $CDF_{tr,2}(\cdot)$  are unknown, then they may be replaced by their estimates. If  $H_0$  is true, then  $CDF_{tr,1}(\cdot)$  is statistically realized  $n_1$  times in the sample (1) for  $j=1$ , and  $CDF_{tr,2}(\cdot)$  is statistically realized  $n_2$  times in the sample (1) for  $j=2$ . Let's shift  $Z^1$  and  $Z^2$  so that both have the same fuzzy sample mean value  $\hat{c}$  as in (3):

$$Z_{c,b}^j = \left\{ (z_{c,1}^j - \mu_k^j), (z_{c,2}^j - \mu_k^j), \dots, (z_{c,n_j}^j - \mu_{n_j}^j) \right\}, \text{ for } j=1,2 \quad (5)$$

where

$$z_{c,k}^j = z_k^j - \hat{c}_j + \hat{c}, \text{ for } k=1, 2, \dots, n_1, \text{ for } j=1,2 \quad (6)$$

Then  $CDF_{tr,1}(\cdot)$  and  $CDF_{tr,2}(\cdot)$  may be intuitively assessed with two different sample step-function approximations:

A) The first sample approximation is the so-called empirical distribution function (ECDF) [Lawless, 2003]:

$$CDF_j(z) = \sum_{\substack{k=1 \\ z_{c,k}^j \leq z}}^{n_j} 1 \Big/ n_j, \text{ for } z \in (-\infty; +\infty), \text{ for } j=1,2 \quad (7)$$

These formulae are the most common approximation of CDF over a rigid sample. It only assumes that the observations in the sample are independent and identically distributed (i.i.d.). ECDF assumes that the r.v. is a discrete one with possible values coinciding with the observations in the rigid sample. The assigned probability to each value is its relative frequency in the sample. The approximation (7) for  $j=1$  is

used by the classical Bootstrap approach, where each observation from the shifted sample (5), for  $j=1$ , has equal chance of being generated, therefore the Bootstrap generation is performed using sampling with replacement from (5), for  $j=1$ . The same applies for the approximation (7) for  $j=2$  and the shifted sample (5), for  $j=2$ .

B) The second sample approximation is the so-called fuzzy empirical distribution functions (FECDF) [Nikolova, Mihaylova, Tenekedjiev, 2015]:

$$CDF_j(z) = \frac{\sum_{k=1}^{n_j} \mu_k^j}{\sum_{k=1}^{n_j} \mu_k^j}, \text{ for } z \in (-\infty; +\infty), \text{ for } j=1,2 \quad (8)$$

Just like the ECDF, the sample approximations (8) use no assumptions for the type of the approximated CDFs except of the standard requirement that the observations are i.i.d. The approximation (8) for  $j=1$  is the cumulative distribution function of a discrete r.v. with possible values coinciding with the different observations in the fuzzy sample (5) for  $j=1$ . The assigned probability to each value is proportional to the sum of the degrees of membership of the instances of that value in the fuzzy sample (5) for  $j=1$ . When using (8) then the Bootstrap approach is modified so that each observation from the shifted sample (5) for  $j=1$  has a probability to be generated proportional to its degree of membership. The same applies for the approximation (8) for  $j=2$  and the shifted fuzzy sample (5), for  $j=2$ .

#### 4. GENERATING SYNTHETIC FZZZY SAMPLES IN A BOOTSTRAP SIMULATION

Assume that in the computer world, the true  $CDF_{w,1}(\cdot)$  and  $CDF_{w,2}(\cdot)$  of the two populations are replaced either by the two ECDF approximations (7), or by the two FECDF approximations (8). Since in the computer world the two populations have equal mean values, then the value of the test statistic would be  $\Delta_{1-2,cw}^{mean} = 0$ . In the computer world, the known  $CDF_1(\cdot)$  and  $CDF_2(\cdot)$  allow to generate  $M$  synthetic pairs of fuzzy samples  $(Z_q^{1,s} - Z_q^{2,s})$ , for  $q=1, 2, \dots, M$ . The synthetic pairs of fuzzy samples can be generated using two intuitive methods ensuring that they resemble the original pair of fuzzy samples  $(Z^1 - Z^2)$ .

A) The first method generates synthetic pairs of fuzzy samples with the same sizes as the original fuzzy samples (1):

$$Z_q^{j,s} = \left\{ \left( z_{1,q}^{j,s} - \mu_{1,q}^{j,s} \right), \left( z_{2,q}^{j,s} - \mu_{2,q}^{j,s} \right), \dots, \left( z_{n_j,q}^{j,s} - \mu_{n_j,q}^{j,s} \right) \right\}, \text{ for } j=1,2 \quad (9)$$

Let's call this generation method for synthetic pairs of fuzzy samples *equal-size generation*. The equal-size generation is used in the classical Bootstrap procedure, and in fact in the entire Monte-Carlo simulation, when the true distributions are considered known [Press et al., 2007].

B) The second method generates synthetic fuzzy samples with approximately the same informational content as the original fuzzy samples (1). The informational content in a

fuzzy sample can be measured with the sum of the degrees of membership of its observations. This method contains three steps.

B1) in the first step, two initial synthetic fuzzy samples with preselected numbers of observations have to be generated, using the known  $CDF_1(\cdot)$  and  $CDF_2(\cdot)$ :

$$Z_{q,ini}^{j,s} = \left\{ \left( z_{1,q}^{j,s} - \mu_{1,q}^{j,s} \right), \left( z_{2,q}^{j,s} - \mu_{2,q}^{j,s} \right), \dots, \left( z_{g_j,q}^{j,s} - \mu_{g_j,q}^{j,s} \right) \right\}, \text{ for } j=1,2 \quad (10)$$

The counts  $g_1$  and  $g_2$  are determined as the minimal sizes which ensure that the initial synthetic fuzzy samples would contain not less information than the original fuzzy samples (1):

$$g_j = \left\lceil \frac{\sum_{k=1}^{n_j} \mu_k^j}{\min\{\mu_1^j, \mu_2^j, \dots, \mu_{n_j}^j\}} \right\rceil, \text{ for } j=1,2 \quad (11)$$

Here  $\lceil x \rceil$  stands for the minimal integer, which is at least  $x$ . In fact  $\lceil \cdot \rceil$  is an operator for upward rounding.

B2) in the second step, the sizes  $n_{1,q}$  and  $n_{2,q}$  of the final synthetic fuzzy samples have to be evaluated in such way that the first  $n_{1,q}$  and  $n_{2,q}$  observations of (10) to have informational content as close as possible to the informational content of the original fuzzy samples (1):

$$n_{j,q} = \arg \min_{i=1,2,\dots,g_j} \left\{ \sum_{k=1}^i \mu_{k,q}^{j,s} - \sum_{k=1}^{n_j} \mu_k^j \right\}, \text{ for } j=1,2 \quad (12)$$

B3) in the third step, the final synthetic fuzzy samples are obtained by cutting down the two generated initial synthetic fuzzy samples to their first  $n_{1,q}$  and  $n_{2,q}$  fuzzy observations:

$$Z_q^{j,s} = \left\{ \left( z_{1,q}^{j,s} - \mu_{1,q}^{j,s} \right), \left( z_{2,q}^{j,s} - \mu_{2,q}^{j,s} \right), \dots, \left( z_{n_{j,q},q}^{j,s} - \mu_{n_{j,q},q}^{j,s} \right) \right\}, \text{ for } j=1,2 \quad (13)$$

Let's call this generation method for synthetic couples of fuzzy samples *quasi-equal-information generation*.

#### 5. ASSESSING PVALUE FOR BOOTSTRAP MEAN VALUE TESTS

Regardless of whether the synthetic samples were derived as in (9) using equal-size generation or as in (13) using quasi-equal-information generation, for the  $q$ -th pair of fuzzy samples it is possible to calculate the synthetic test statistic  $\Delta_{1-2,q}^{mean,s}$  using (4), which is one realization of the sample

numerical characteristic in the computer world  $\Delta_{1-2}^{mean,s}$ :

$$\Delta_{1-2,q}^{mean,s} = \frac{\sum_{k=1}^{n_{1,g}} \mu_{k,q}^{1,s} z_{k,q}^{1,s}}{\sum_{k=1}^{n_{1,g}} \mu_{k,q}^{1,s}} - \frac{\sum_{k=1}^{n_{2,g}} \mu_{k,q}^{2,s} z_{k,q}^{2,s}}{\sum_{k=1}^{n_{2,g}} \mu_{k,q}^{2,s}} \quad (14)$$

In (14),  $n_{1,g}$  and  $n_{2,g}$  are the generalized sizes of the synthetic fuzzy samples in the  $q^{\text{th}}$  synthetic couple, for  $j=1,2$ :

$$n_{j,g} = \begin{cases} n_j & \text{for equal-size generation} \\ n_{j,q} & \text{for quasi-equal information generation} \end{cases} \quad (15)$$

After  $M$  pseudo-realities, one can get a synthetic sample  $S_{mean}$  containing  $M$  number of instances of the random variable  $\Delta_{1-2}^{mean,s}$ :

$$S_{mean} = \{\Delta_{1-2,1}^{mean,s}, \Delta_{1-2,2}^{mean,s}, \dots, \Delta_{1-2,M}^{mean,s}\} \quad (16)$$

If  $M$  is high (for example 100000), then the cumulative distribution of the random variable  $\Delta_{1-2}^{mean,s}$  may be constructed with high precision as ECDF over (16):

$$CDF_{\Delta_{1-2}^{mean,s}}(\Delta) = \frac{1}{M} \sum_{\substack{q=1 \\ \Delta_{1-2,q}^{mean,s} \leq \Delta}}^M 1, \text{ for } k_u \in (-\infty; +\infty) \quad (17)$$

The difference  $\Delta_{1-2}^{mean,s} - \Delta_{1-2,cw}^{mean} = \Delta_{1-2}^{mean,s} - 0 = \Delta_{1-2}^{mean,s}$  is an r.v. called  $root_{cw}$ . The main assumption in the Bootstrap technique is that the distribution of  $root_{cw}$  in the computer world is the same as the distribution of  $root$  in the real world. Since in that case,  $root$  coincides with  $\Delta_{1-2}^{mean}$ , and  $root_{cw}$  coincides with  $\Delta_{1-2}^{mean,s}$ , then the necessary distribution of the sample numerical characteristic  $\Delta_{1-2}^{mean}$  coincides with the constructed distribution of  $\Delta_{1-2}^{mean,s}$  in (17). In fact, it is necessary to calculate  $p_{value}$  of the three two-tail and the two one-tail tests, not the distribution itself. Since  $M$  is a large number, then  $p_{value}$  may be very well assessed using frequentist approaches [Xie, Singh, 2013].

In the case of two-tail test, using the frequentist interpretation to probabilities,  $p_{value}$  is the relative count of synthetic test statistics  $\Delta_{1-2,q}^{mean,s}$ , whose modulae are not less than the modulus of the observed test statistic  $\Delta_{1-2,r}^{mean}$ :

$$p_{value} = \frac{1}{M} \left( \sum_{q=1}^M 1 + \sum_{q=1}^M 1 \right) \quad (18)$$

$$\left( \Delta_{1-2,q}^{mean,s} \geq |\Delta_{1-2,r}^{mean}| \mid \Delta_{1-2,q}^{mean,s} \leq -|\Delta_{1-2,r}^{mean}| \right)$$

In the case of one-tail test of first type,  $p_{value}$  shall be the relative count of synthetic mean value differences that are not less than the modulus of the observed mean test statistic  $\Delta_{1-2,r}^{mean}$ :

$$p_{value} = \frac{1}{M} \left( \sum_{q=1}^M 1 \right) \quad (19)$$

$$\left( \Delta_{1-2,q}^{mean,s} \geq |\Delta_{1-2,r}^{mean}| \right)$$

In the case of one-tail test of second type,  $p_{value}$  shall be the relative count of synthetic mean value differences that are not greater than the opposite of the modulus of the observed mean test statistic  $\Delta_{1-2,r}^{mean}$ :

$$p_{value} = \frac{1}{M} \left( \sum_{q=1}^M 1 \right) \quad (20)$$

$$\left( \Delta_{1-2,q}^{mean,s} \leq -|\Delta_{1-2,r}^{mean}| \right)$$

In fact, the one-tail test of first type applies when  $\hat{c}_1 > \hat{c}_2$  and then the condition in the sum of (19) becomes  $\Delta_{1-2,q}^{mean,s} \geq \Delta_{1-2,r}^{mean}$ , because  $\Delta_{1-2,q}^{mean,s} > 0$ . In the same way, the one-tail test of second type is performed only when  $\hat{c}_1 < \hat{c}_2$  and then the condition in the sum of (20) becomes  $\Delta_{1-2,q}^{mean,s} \leq \Delta_{1-2,r}^{mean}$ , because  $\Delta_{1-2,q}^{mean,s} < 0$ .

## 6. ALGORITHMS TO CALCULATE PVALUE

This section offers an algorithm to calculate the  $p_{value}$  for the mean value difference test over fuzzy samples. This algorithm had eight modifications that follow from the method to generate the synthetic pairs of fuzzy samples, the type of the conditional sample distributions derived from the shifted fuzzy samples, and the type of the statistical test performed (one-tail or two-tail test).

### Bootstrap algorithm for estimation of $p_{value}$ for the mean value difference test over fuzzy samples

1. Select the generation method for synthetic pairs of fuzzy samples:

$$Fgen = \begin{cases} 0 & \text{for 'quasi-equal-information generation'} \\ 1 & \text{for 'equal-size generation'} \end{cases}$$

2. Select the type of the conditional sample distributions derived from the shifted fuzzy samples:

$$Fdist = \begin{cases} 0 & \text{for ECDC} \\ 1 & \text{for FECDC} \end{cases}$$

3. Select the type of the statistical test performed:

$$Ftail = \begin{cases} 0 & \text{for two-tail test} \\ 1 & \text{for one-tail test} \end{cases}$$

4. Select the count of pseudo-realities  $M$

5. Calculate the fuzzy sample mean  $\hat{c}_1$  of Fuzzy Sample 1 using (2) for  $j=1$ .

6. Calculate the fuzzy sample mean  $\hat{c}_2$  of Fuzzy Sample 2 using (2) for  $j=2$ .

7. Calculate the fuzzy sample mean  $\hat{c}$  of the unified fuzzy sample using (3).

8. Estimate the actual realization of the test statistic  $\Delta_{1-2,r}^{mean}$  according to (4).

9. Form the first shifted fuzzy sample  $Z_{c,b}^1$  according to (5) using (6), for  $j=1$ , to shift the observations in Fuzzy Sample 1.

10. Form the second shifted fuzzy sample  $Z_{c,b}^2$  according to (5) using (6), for  $j=2$ , to shift the observations in Fuzzy Sample 2.

11. If  $Fgen$  is 0, then determine the fixed sizes of the initial synthetic fuzzy samples:

a) Calculate the size  $g_1$  of the first initial synthetic samples  $Z_{q,ini}^{1,s}$  according to (11), for  $j=1$

b) Calculate the size  $g_2$  of the second initial synthetic samples  $Z_{q,ini}^{2,s}$  according to (11), for  $j=2$

12. Initialize pseudo-reality cycle:  $q=1$ ;

13. Form the synthetic fuzzy sample  $Z_q^{1,s}$ :

- a) If  $Fgen$  is 0, then go to step 13h;
- b) Initialize first synthetic observation cycle:  $k=1$ ;
- c) If  $Fdist$  is 0, then draw the  $k^{th}$  observation of  $Z_q^{1,s}$  from the first shifted fuzzy sample  $Z_{c,b}^1$  by generation according to (7), for  $j=1$ :
  - \* generate a pseudo-random number  $rand \in [0;1)$  from an even distribution
  - \* find the number in  $Z_{c,b}^1$  of the  $k^{th}$  observation of  $Z_q^{1,s}$ :  $numt = \lceil rand \times n_1 \rceil$
  - \* remember the  $k^{th}$  observation of  $Z_q^{1,s}$ :  $(z_{k,q}^{1,s} - \mu_{k,q}^{1,s}) = (z_{c,numt}^1 - \mu_{numt}^1)$
- d) If  $Fdist$  is 1, then draw the  $k^{th}$  observation of  $Z_q^{1,s}$  from the first shifted fuzzy sample  $Z_{c,b}^1$  by generation according to (8), for  $j=1$ :
  - \* generate a pseudo-random number  $rand \in [0;1)$  from an even distribution
  - \* find the number in  $Z_{c,b}^1$  of the  $k^{th}$  observation of  $Z_q^{1,s}$ :  $numt$  is the minimal  $i$  from the set  $\{1,2,\dots, n_1\}$  which satisfies the condition  $\left(\sum_{k=1}^i \mu_k^1\right) > rand$
  - \* remember the  $k^{th}$  observation of  $Z_q^{1,s}$ :  $(z_{k,q}^{1,s} - \mu_{k,q}^{1,s}) = (z_{c,numt}^1 - \mu_{numt}^1)$
- e) Move to the next observation:  $k=k+1$ ;
- f) Check whether there are enough generated observations for  $Z_q^{1,s}$ : if  $k \leq n_1$ , then go to step 13c;
- g) Form the synthetic fuzzy sample  $Z_q^{1,s}$  according to (9), for  $j=1$ , and go to step 14;
- h) Initialize first initial synthetic observation cycle:  $k=1$ ;
- i) If  $Fdist$  is 0, then draw the  $k^{th}$  observation of  $Z_{q,ini}^{1,s}$  from the first shifted fuzzy sample  $Z_{c,b}^1$  by generation according to (7), for  $j=1$ :
  - \* generate a pseudo-random number  $rand \in [0;1)$  from an even distribution
  - \* find the number in  $Z_{c,b}^1$  of the  $k^{th}$  observation of  $Z_q^{1,s}$ :  $numt = \lceil rand \times n_1 \rceil$
  - \* remember the  $k^{th}$  observation of  $Z_{q,ini}^{1,s}$ :  $(z_{k,q}^{1,s} - \mu_{k,q}^{1,s}) = (z_{c,numt}^1 - \mu_{numt}^1)$
- j) If  $Fdist$  is 1, then draw the  $k^{th}$  observation of  $Z_{q,ini}^{1,s}$  from the first shifted fuzzy sample  $Z_{c,b}^1$  by generation according to (8), for  $j=1$ :
  - \* generate a pseudo-random number  $rand \in [0;1)$  from an even distribution

\* find the number in  $Z_{c,b}^1$  of the  $k^{th}$  observation of  $Z_{q,ini}^{1,s}$ :  $numt$  is the minimal  $i$  from the set  $\{1,2,\dots, n_1\}$

which satisfies the condition  $\left(\sum_{k=1}^i \mu_k^1\right) > rand$

\* remember the  $k^{th}$  observation of  $Z_q^{1,s}$ :  $(z_{k,q}^{1,s} - \mu_{k,q}^{1,s}) = (z_{c,numt}^1 - \mu_{numt}^1)$

k) Move to the next observation:  $k=k+1$ ;

l) Check whether there are enough generated observations for  $Z_{q,ini}^{1,s}$ : if  $k \leq g_1$ , then go to step 13i;

m) Form the initial synthetic fuzzy sample  $Z_{q,ini}^{1,s}$  according to (10), for  $j=1$ ;

n) Calculate the size  $n_{1,q}$  of the final synthetic fuzzy sample  $Z_q^{1,s}$  according to (12), for  $j=1$ ;

o) Form the final synthetic fuzzy sample  $Z_q^{1,s}$  according to (13), for  $j=1$ , by cutting down  $Z_{q,ini}^{1,s}$  to its first  $n_{1,q}$  fuzzy observations;

14. Form the synthetic fuzzy sample  $Z_q^{2,s}$ :

- a) If  $Fgen$  is 0, then go to step 14h;
- b) Initialize second synthetic observation cycle:  $k=1$ ;
- c) If  $Fdist$  is 0, then draw the  $k^{th}$  observation of  $Z_q^{2,s}$  from the second shifted fuzzy sample  $Z_{c,b}^2$  by generation according to (7), for  $j=2$ :
  - \* generate a pseudo-random number  $rand \in [0;1)$  from an even distribution
  - \* find the number in  $Z_{c,b}^2$  of the  $k^{th}$  observation of  $Z_q^{2,s}$ :  $numt = \lceil rand \times n_2 \rceil$
  - \* remember the  $k^{th}$  observation of  $Z_q^{2,s}$ :  $(z_{k,q}^{2,s} - \mu_{k,q}^{2,s}) = (z_{c,numt}^2 - \mu_{numt}^2)$
- d) If  $Fdist$  is 1, then draw the  $k^{th}$  observation of  $Z_q^{2,s}$  from the second shifted fuzzy sample  $Z_{c,b}^2$  by generation according to (8), for  $j=2$ :
  - \* generate a pseudo-random number  $rand \in [0;1)$  from an even distribution
  - \* find the number in  $Z_{c,b}^2$  of the  $k^{th}$  observation of  $Z_q^{2,s}$ :  $numt$  is the minimal  $i$  from the set  $\{1,2,\dots, n_2\}$  which satisfies the condition  $\left(\sum_{k=1}^i \mu_k^2\right) > rand$
  - \* remember the  $k^{th}$  observation of  $Z_q^{2,s}$ :  $(z_{k,q}^{2,s} - \mu_{k,q}^{2,s}) = (z_{c,numt}^2 - \mu_{numt}^2)$
- e) Move to the next observation:  $k=k+1$ ;
- f) Check whether there are enough generated observations for  $Z_q^{2,s}$ : if  $k \leq n_2$ , then go to step 14c;
- g) Form the synthetic fuzzy sample  $Z_q^{2,s}$  according to (9), for  $j=2$ , and go to step 15;

- h) Initialize second initial synthetic observation cycle:  $k=1$ ;
- i) If  $Fdist$  is 0, then draw the  $k^{th}$  observation of  $Z_{q,ini}^{2,s}$  from the second shifted fuzzy sample  $Z_{c,b}^2$  by generation according to (7), for  $j=2$ :
- \* generate a pseudo-random number  $rand \in [0;1)$  from an even distribution
  - \* find the number in  $Z_{c,b}^2$  of the  $k^{th}$  observation of  $Z_q^{2,s}$ :  $numt = \lceil rand \times n_2 \rceil$
  - \* remember the  $k^{th}$  observation of  $Z_{q,ini}^{2,s}$ :  $(z_{k,q}^{2,s} - \mu_{k,q}^{2,s}) = (z_{c,numt}^2 - \mu_{numt}^2)$
- j) If  $Fdist$  is 1, then draw the  $k^{th}$  observation of  $Z_{q,ini}^{2,s}$  from the second shifted fuzzy sample  $Z_{c,b}^2$  by generation according to (8), for  $j=2$ :
- \* generate a pseudo-random number  $rand \in [0;1)$  from an even distribution
  - \* find the number in  $Z_{c,b}^2$  of the  $k^{th}$  observation of  $Z_{q,ini}^{2,s}$ :  $numt$  is the minimal  $i$  from the set  $\{1, 2, \dots, n_2\}$  which satisfies the condition  $\left( \sum_{k=1}^i \mu_k^2 \right) > rand$
  - \* remember the  $k^{th}$  observation of  $Z_q^{2,s}$ :  $(z_{k,q}^{2,s} - \mu_{k,q}^{2,s}) = (z_{c,numt}^2 - \mu_{numt}^2)$
- k) Move to the next observation:  $k=k+1$ ;
- l) Check whether there are enough generated observations for  $Z_{q,ini}^{2,s}$ : if  $k \leq g_2$ , then go to step 14i;
- m) Form the initial synthetic fuzzy sample  $Z_{q,ini}^{2,s}$  according to (10), for  $j=2$ ;
- n) Calculate the size  $n_{2,q}$  of the final synthetic fuzzy sample  $Z_q^{2,s}$  according to (12), for  $j=2$ ;
- o) Form the final synthetic fuzzy sample  $Z_q^{2,s}$  according to (13), for  $j=2$ , by cutting down  $Z_{q,ini}^{2,s}$  to its first  $n_{2,q}$  fuzzy observations.
15. Calculate the generalized number  $n_{1,g}$  of the fuzzy observations in  $Z_q^{1,s}$  according to (15), for  $j=1$ ;
16. Calculate the generalized number  $n_{2,g}$  of the fuzzy observations in  $Z_q^{2,s}$  according to (15), for  $j=2$ ;
17. Calculate the test statistic  $\Delta_{1-2,q}^{mean,s}$  in the  $q^{th}$  pseudo-reality according to (14);
18. Move to the next pseudo-reality:  $q=q+1$ ;
19. Check for end of simulation: if  $q \leq M$ , then go to step 13;
20. Form the synthetic sample  $S_{mean}$  containing  $M$  instances of the random variable  $\Delta_{1-2}^{mean,s}$  according to (16);
21. If  $Ftail$  is 0 then calculate  $p_{value}$  according to (18). Define the alternative hypothesis  $H_1$  "the mean value of Population 1 is different from the mean value of Population 2:  $c_1 \neq c_2$ ";

22. If  $Ftail$  is 1 and  $\hat{c}_1 > \hat{c}_2$  then calculate  $p_{value}$  according to (19). Define the alternative hypothesis  $H_1$  "the mean value of Population 1 is greater than the mean value of Population 2:  $c_1 > c_2$ ";
23. If  $Ftail$  is 1 and  $\hat{c}_1 < \hat{c}_2$  then calculate  $p_{value}$  according to (20). Define the alternative hypothesis  $H_1$  "the mean value of Population 1 is smaller than the mean value of Population 2:  $c_1 < c_2$ ";
24. If  $Ftail$  is 1 and  $\hat{c}_1 = \hat{c}_2$ , then  $p_{value}$  is not defined, and  $H_1$  is not defined as well.

## 7. NUMERICAL EXPERIMENTATIONS

**Numerical example 1:** Assume there are two fuzzy samples  $Z^1 = \{(15.25-1), (17.16-1), (11.55-1), (17.28-1), (23.88-1), (20.11-1)\}$  and  $Z^2 = \{(15.19-1), (12.28-0.5), (31.38-0.7), (18.80-1), (15.30-0.6), (19.02-0.8), (18.97-1), (21.28-1)\}$ . The rigid samples containing the same observations are discussed in [Nikolova, Mihaylova, Tenekedjiev, 2013]. The same fuzzy samples are used in [Nikolova, Mihaylova, Tenekedjiev, 2015] to test the hypothesis for equal distributions of the parameter in the two populations. The same problem is solved with the rigid samples containing the same observations in [Nikolova, Mihaylova, Tenekedjiev, 2013]. Here the samples are used to test the hypothesis for equal mean values of the parameter in the two populations. The test statistic  $\Delta_{1-2,r}^{mean}$  is -1.665 according to (4). Each of the four modifications of the proposed algorithm are realized with  $M=1000$  pseudo-realities. The corresponding Bootstrap generated CDFs of the difference of means statistic are shown on Fig.1. The one-tail and two-tail tests  $p_{value}$  for each of the mean value difference test modifications are: a) 0.243 and 0.477 for Bootstrap with quasi-equal-information generation using ECDF; b) 0.250 and 0.470 for Bootstrap with quasi-equal-information generation using FECDF; c) 0.230 and 0.464 for Bootstrap with equal-size generation using ECDF; d) 0.223 and 0.433 for Bootstrap with equal-size generation using FECDF. Obviously, the data in the fuzzy samples does not contradict the null hypothesis for any reasonable significance level.

In fact, the observations in the two samples are generated from equal continuous distributions of the two populations. The unified distribution is a mixture of 40% normal distribution with a mathematical expectation 14 and variance 16, and 60%  $\chi^2$  with two degrees of freedom, with expectation 19 and variance 16. Since here,  $CDF_{tr}(z)$  is known, then the true distribution of the difference of means statistic is derived using Monte Carlo simulation with 1000 hypothetic samples. In the  $q^{th}$  pseudo-reality, there are respectively 6 and 8 observations generated from  $CDF_{tr}(z)$ . Each of the 14 observations has an associated degree of membership to the respective population from the given  $Z^1$  and  $Z^2$ . In that way, a synthetic pair in the analyzed pseudo reality is formed from fuzzy samples  $(Z_q^{1,MK} - Z_q^{2,MK})$ , for  $q=1, 2, \dots, 1000$ . Using the  $q^{th}$  pair of samples it is possible to calculate the realization  $\Delta_{1-2,q}^{mean,MC}$  of the difference of means

statistic in a pseudo-reality. Based on the resulting pseudo-realizations  $\Delta_{1-2,1}^{mean,MC}, \Delta_{1-2,2}^{mean,MC}, K, \Delta_{1-2,1000}^{mean,MC}$  it is possible to construct the CDF of the difference of means statistic using Monte Carlo simulation. The latter is presented on each of the four graphics on Figure 1. It is obvious that due to the small number of fuzzy samples, the approximation of the distribution of the difference of means statistic is not very close to the real Monte Carlo approximation. In the same time, there is no substantial difference between the results from the four versions of the proposed algorithm.

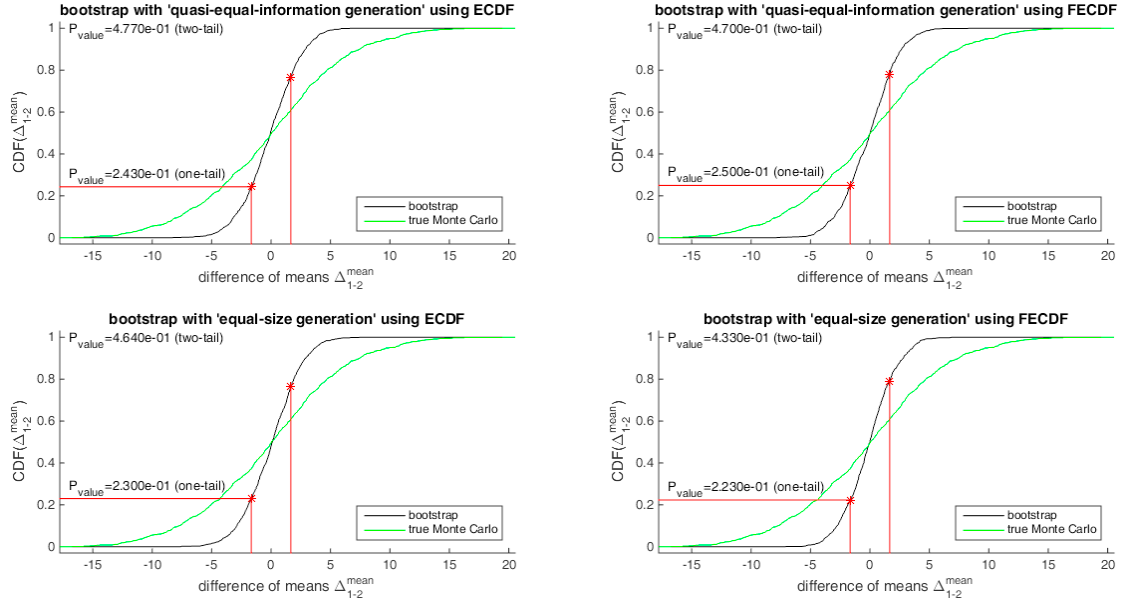


Fig. 1. Bootstrap generated CDFs of the difference of means statistic for the fuzzy samples in the Numerical example 1, along with the true Monte-Carlo empirical CDF of the same data

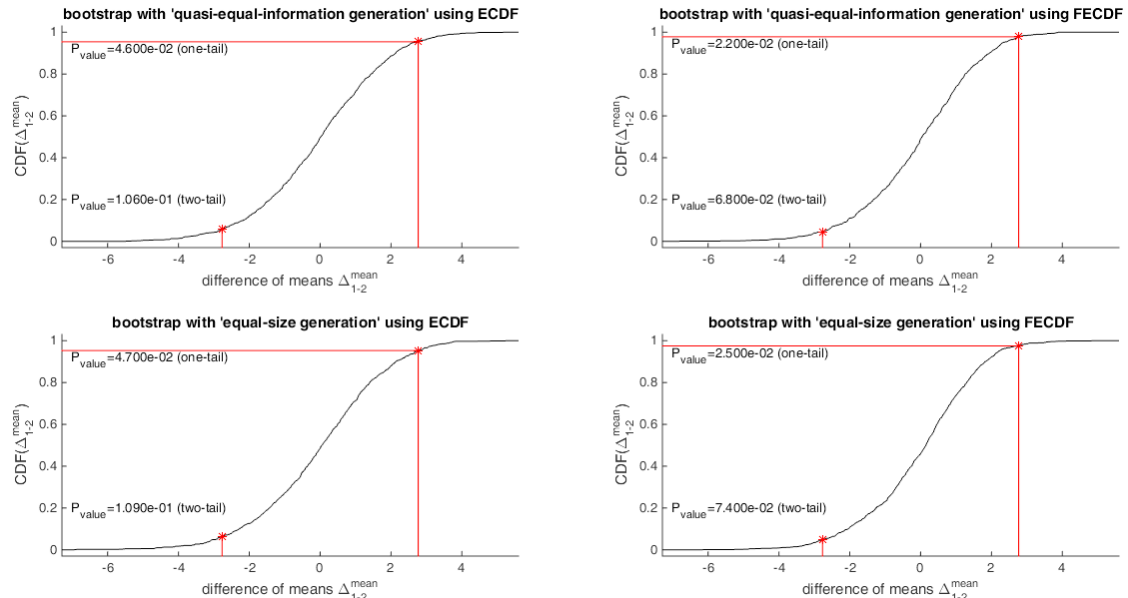


Fig. 2. Bootstrap generated CDFs of the difference of means statistic for the fuzzy samples in Numerical example 2

The one-tail and two-tail tests  $p_{value}$  for each of the mean value difference test modifications are: a) 0.046 and 0.106 for Bootstrap with quasi-equal-information generation using

**Numerical example 2:** Let  $Z^1 = \{(21.38-1), (22.01-1), (20.14-1), (21.05-1), (24.25-1), (23.00-1)\}$  and  $Z^2$  be as in Numerical Example 1. The test statistic calculated using (4) is  $\Delta_{1-2,r}^{mean} = 2.935$ . Each of the four modifications of the proposed algorithm is realized with  $M=1000$  pseudo-realities and the corresponding Bootstrap generated CDFs of the difference of means statistic are shown on Fig. 2.

ECDF; b) 0.022 and 0.068 for Bootstrap with quasi-equal-information generation using FECDF; c) 0.047 and 0.109 for Bootstrap with equal-size generation using ECDF; d) 0.025 and 0.074 for Bootstrap with equal-size generation using

FECDF. At 5% significance level, the null hypothesis has to be rejected by each of the four one-tail modifications of the Bootstrap mean value difference test with fuzzy samples. However, at the same significance level, the null hypothesis fails to be rejected by each of the four two-tail modifications of the Bootstrap mean value difference test with fuzzy samples. As a whole, there is no recognizable difference between the results of the four modifications of the proposed algorithm, in the same way as in Numerical example 1.

## 8. CONCLUSIONS

There are practical cases, where fuzzy samples arise, as described in [Viertl, 2011]. For example, if we compare two subpopulations of two different populations, then sometimes the membership of a given observation to the subpopulation is not defined trivially but it is calculated from a given classifier. The degree of certainty in the solution of the classifier for a particular observation may be interpreted as a degree of membership to the sample of the subpopulation. This setup allows using any type of classifier: probabilistic, metric, neural network, even subjective. Another practical implementation arises when for some objects there is more than one replica of complex measurements. In medical analyses, for example, a given parameter may be measured in  $t$  spatial points (e.g. measurements of different parts of a thrombus) for a given patient. Then the degree of membership to the sample for such measurements may be accepted as  $1/t$  so that to provide equal weight for each patient. This shows that the interpretation of the degree of membership is problem-dependent.

In this paper we used fuzzy samples of 1D continuous parameter, where a degree of membership to the corresponding population was assigned to each i.i.d. observation. Then the objective was to test the null hypothesis  $H_0$  that the mean values in the two populations were equal. Bootstrap was employed to approximate the CDF of the difference between the sample means, if  $H_0$  was true. An algorithm with eight modifications was developed to calculate the  $p_{value}$  of one-tail and two-tail statistical tests for equality of means. The algorithm was realized in original software in MATLAB R2015a, that is available free of charge upon request from the authors. The procedures and the developed software were demonstrated in two numerical examples.

The modifications of the proposed algorithm differ by the method to generate the synthetic fuzzy sample, by the type of the conditional sample distributions derived from the shifted fuzzy samples, and by the type of the statistical test performed (one-tail or two-tail test). The type of test is problem-specific. Future experiments can show which method to generate the synthetic fuzzy sample should be employed, but both that were proposed look equally promising at the stage of theoretical discussion.

Choosing the type of the conditional sample distributions derived from the shifted fuzzy samples deserves some

attention. The CDF of any population can be better approximated by the FECDF of a fuzzy sample, compared to approximation by the ECDF of the same sample. At first glance it appears that the generation of the synthetic samples in the different pseudo-realities should utilize the FECDF, rather than the ECDF of the unified fuzzy sample. The last statement is, however, flawed. A low degree of membership of a specific fuzzy observation, included in a synthetic fuzzy sample, ensures that it is going to influence less the sample FECDF of the parameter in the pseudo-reality. Using the FECDF of the unified fuzzy sample for generation of the synthetic samples ensures that this synthetic observation has even less chance to be included in the synthetic fuzzy sample. So the low degree of membership is in a way double-accounted. That is why, it can be expected that the modifications, which generate the synthetic fuzzy samples from the ECDF of the unified fuzzy sample, would produce better results than the corresponding ones, which utilize the FECDF of the unified fuzzy sample for generation. However, the work [Nikolova, Mihaylova, Tenekedjiev, 2015] offered initial numerical results from examples in a similar content, which did not support this claim, and produced similar results.

New extensive numerical simulations and new theoretical advances are needed to clarify which of the eight modifications should be trusted. It is quite possible to have problem- and/or interpretation-dependent answer.

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